

Answer Scheme Final Exam
FSPC0034 Chemistry II
Semester 3 Year 2019/2020

Foundation Programme SPACE UTM

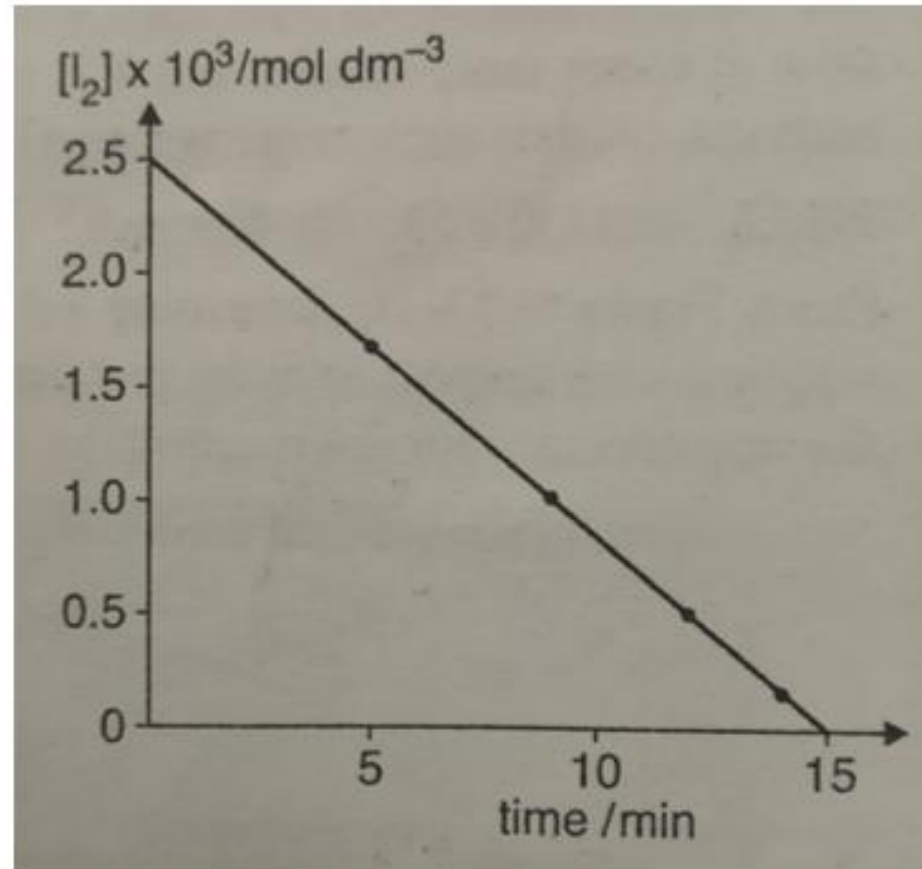
Question 1

(a)	(i)	$2 \text{ Al (s)} + 3/2 \text{ O}_2 \text{ (g)} \rightarrow \text{Al}_2\text{O}_3 \text{ (s)}$ $2\text{Al (g)} \rightarrow 2\text{Al (s)} \text{ (flip + x2)}$ $3 \text{ O (g)} \rightarrow 3/2 \text{ O}_2 \text{ (g)} \text{ (flip + x3)}$ $2 \text{ Al}^{3+} \text{ (g)} + 6 \text{ e}^- \rightarrow 2 \text{ Al (g)} \text{ (flip + x2)}$ $\text{O}^{2-} \text{ (g)} \rightarrow \text{O (g)} + 2 \text{ e}^- \text{ (flip + x3)}$ $\text{Total (lattice energy)}$	$\begin{array}{r} -1\,675.7 \\ -652 \\ -1\,494 \\ -10\,278 \\ -2\,532 \\ \hline -16\,631.7 \end{array}$
	(ii)	Stable. Negative enthalpy indicates <u>exothermic process</u> which resulted in a <u>stable structure</u>.	
(b)		$m = 9.55\text{g} + 60\text{g}$ $\Delta T = 18.4^\circ\text{C} - 23^\circ\text{C}$ $q = m \times s \times \Delta T + C\Delta T$ $= (9.55\text{g} + 60\text{g}) \times 4.186 \text{ J}^\circ\text{C}^{-1}\text{g}^{-1} \times (18.4 - 23) + 22.95 \text{ J}^\circ\text{C}^{-1} \times (18.4 - 23)$ $= -1\,444.8 \text{ J}$ Endothermic	

Question 2

(a)

(i)



(ii)

Order of reaction: zero order

Justification: The plot of concentration of iodine against time is a straight line. Rate of reaction with respect to iodine is a constant.

Question 2

(b)	(i)	$2\text{NO (g)} + \text{Cl}_2 \text{(g)} \rightarrow 2\text{ClNO (g)}$
	(ii)	Step 1: $\text{NO (g)} + \text{Cl}_2 \text{(g)} \rightarrow \text{ClNO (g)} + \text{Cl} \cdot \text{(g)}$
		Step 2: $\text{NO (g)} + \text{Cl} \cdot \text{(g)} \rightarrow \text{ClNO (g)}$
	(iii)	Step 1 is the rate-determining step.
		In Step 1, covalent bonds of Cl_2 are broken, which requires energy. This is the slow step.

Question 3

(a)	(i)	At equilibrium : $K_p = (P_{H_2})^4 / (P_{H_2O})^4$ $K_p = (319)^4 / (15.2)^4$ $K_p = 19.4 \times 10^4$
	(ii)	$P_T = P(H_2O) + P(H_2) = 100 \text{ atm}$ Assume that : $P(H_2O) = x$, then $P(H_2) = 100 - x$ At Equilibrium : $K_p = (P_{H_2})^4 / (P_{H_2O})^4$ $K_p = (100-x)^4 / (x)^4 = 19.4 \times 10^4$ $100 = 21.987x$ $x = 4.55 \text{ atm}$ ❖ $P(H_2O) = x = 4.55 \text{ atm}$ ❖ $P(H_2) = 100 - x = 100 - 4.55 = 95.45 \text{ atm}$

Question 3

(b)	(i)	[HI] : increase because we favour forward reaction
	(ii)	Equilibrium position shift to the right (favour forward reaction) to absorb heat
	(iii)	K_p : increase since we favour forward reaction
(c)		$\text{SO}_2 (\text{g}) + \frac{1}{2} \text{O}_2 (\text{g}) \rightleftharpoons \text{SO}_3 (\text{g})$ is equivalent to $\frac{1}{2} \times [2 \text{SO}_2 (\text{g}) + \text{O}_2 (\text{g}) \rightleftharpoons 2 \text{SO}_3 (\text{g})]$ So, $K'_p = [K_p]^{1/2}$ $K'_p = [1.3 \times 10^4]^{1/2}$ $K'_p = 114$

Question 4

(a)	(i)	$\text{HCOOH (aq)} + \text{H}_2\text{O (l)} \rightleftharpoons \text{H}_3\text{O}^+ \text{ (aq)} + \text{HCOO}^- \text{ (aq)}.$				
		I	0.534 mol L ⁻¹	—	0	0
		C	—x	—	+x	+x
		E	0.534—x	—	x	x
		$K_a = \frac{[\text{HCOO}^-][\text{H}_3\text{O}^+]}{[\text{HCOOH}]} = 1.8 \times 10^{-4}$ $= \frac{(x)(x)}{0.534-x} = 1.8 \times 10^{-4} \text{ (since } K_a < 10^2, \text{ it can be neglect)}$ $x^2 = (1.8 \times 10^{-4})(0.534) = 9.612 \times 10^{-5}$ $x = 9.804 \times 10^{-3}$ $[\text{H}_3\text{O}^+] = x = 9.804 \times 10^{-3} \text{ mol L}^{-1}$ $\text{pH} = -\log [\text{H}_3\text{O}^+] = -\log (9.804 \times 10^{-3}) = 2.01$				
	(ii)	$\frac{[\text{HCOO}^-]}{[\text{HCOOH}]} \times 100\%$ $\frac{9.804 \times 10^{-3}}{0.534} \times 100\% = 1.84\%$				

Question 4

(b)	(i)	Total volume = 800 ml (CH ₃ COOH) + 200 ml (NaOH) = 1000 ml or 1L																								
		<table><tr><td></td><td colspan="4">CH₃COOH (aq) + NaOH (aq) ⇌ CH₃COONa (aq) + H₂O (l)</td></tr><tr><td>I</td><td>= 0.8 ml × 0.10 mol L⁻¹ = 0.08 mol</td><td>= 0.2 ml × 0.10 mol L⁻¹ = 0.02 mol</td><td>0</td><td>-</td></tr><tr><td>C</td><td>-0.02 mol</td><td>-0.02 mol</td><td>+0.02 mol</td><td>-</td></tr><tr><td>E</td><td>0.06 mol</td><td>0 mol</td><td>+0.02 mol</td><td>-</td></tr></table>						CH ₃ COOH (aq) + NaOH (aq) ⇌ CH ₃ COONa (aq) + H ₂ O (l)				I	= 0.8 ml × 0.10 mol L ⁻¹ = 0.08 mol	= 0.2 ml × 0.10 mol L ⁻¹ = 0.02 mol	0	-	C	-0.02 mol	-0.02 mol	+0.02 mol	-	E	0.06 mol	0 mol	+0.02 mol	-
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		<table><tr><td>[CH₃COOH] left</td><td>= 0.06 mol/ 1L</td><td>= 0.06 mol L⁻¹</td></tr><tr><td>[CH₃COONa]</td><td>= 0.02 mol/ 1L</td><td>= 0.02 mol L⁻¹</td></tr></table>					[CH ₃ COOH] left	= 0.06 mol/ 1L	= 0.06 mol L ⁻¹	[CH ₃ COONa]	= 0.02 mol/ 1L	= 0.02 mol L ⁻¹														
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[CH ₃ COONa]	= 0.02 mol/ 1L	= 0.02 mol L ⁻¹																								
	(ii)	pH = pKa + log ([base]÷[acid]) = 4.74 + log (0.02÷0.06) = 4.74 - 0.477 = 4.263																								

Question 4

(iii)	No. of mole of sodium ethanoate = 0.02 mol MW of CH₃COONa = 82 g mol⁻¹ Mass of sodium ethanoate, CH₃COONa = 82 g mol⁻¹ × 0.02 mol = 1.64 g or
	No. of mole of ethanoic acid = 0.08 mol MW of CH₃COOH = 60 g mol⁻¹ Mass of sodium ethanoate, CH₃COOH = 60 g mol⁻¹ × 0.08 mol = 4.8 g
(iv)	CH₃COO⁻ + H₂O ⇌ CH₃COOH + OH⁻ Ethanoic acid (CH₃COOH) is a weak acid. The pH is higher than 7 because the presence of OH⁻ ions which are strongly alkaline.

Question 5



← In acidic medium

Mark's given for correct/balanced equation

In basic medium

the number of OH equivalent to the number of H⁺



Mark's given for correct coefficients of OH⁻ on both sides (left/right)



Question 5

(b)

Point 1:

Competing ions at anode (oxidation) : OH^- and SO_4^{2-}

Competing ions at cathode (reduction): H^+ and Na^+

Point 2:

Ions selectively discharge at

Anode: OH^-

Cathode: H^+

Point 3:

OH^- is lower than SO_4^{2-} in the electrochemical series (SO_4^{2-} ions are not oxidized at the anode.)

H^+ is lower than Na^+ in the electrochemical series (The Na^+ ions are not reduced at the cathode)

Point 4:

Both the oxidation and reduction reactions involve only water molecules.

Anode (oxidation): $2\text{H}_2\text{O}(\text{l}) \rightarrow \text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) + 4\text{e}^-$

Cathode(reduction): $4\text{H}_2\text{O}(\text{l}) + 4\text{e}^- \rightarrow 2\text{H}_2(\text{g}) + 4\text{OH}^-(\text{aq})$

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Point 5:

The overall reaction:

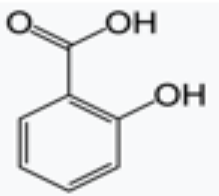
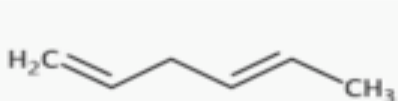
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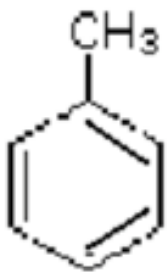
$4\text{H}_2\text{O}(\text{l}) + 4\text{e}^- \rightarrow 2\text{H}_2(\text{g}) + 4\text{OH}^-(\text{aq})$

$6\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{H}_2(\text{g}) + 4\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$

$2\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{H}_2(\text{g}) + \text{O}_2(\text{g})$

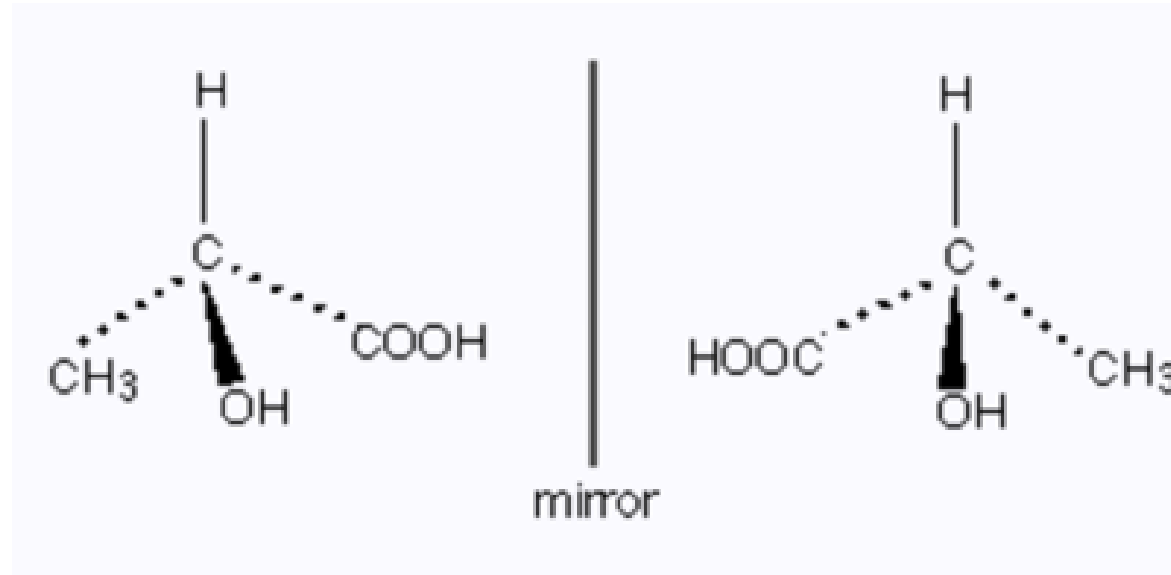
Question 6

(a)	(i)	 General formula: C_nH_{2n+2}
	(ii)	 General formula: $C_nH_{2n+2}O$
	(iii)	 General formula: C_nH_{2n+2}
	(iv)	 General formula: C_nH_{2n}

		 General formula: C_nH_{2n}
(b)	(i)	2-methylhexan-2-ol OR 2-methyl-2-hexanol alcohol
	(ii)	Cyclopentene alkene
	(iii)	3-chloro-4-methylhept-3-ene alkene

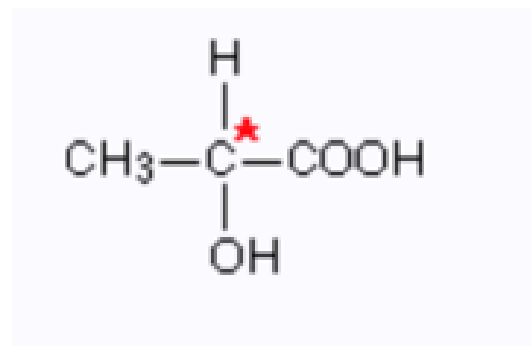
Question 6

(c)



Correct isomer pair

Correct labelling of chiral carbon (*)



(d)	(i)	homopolymer
	(ii)	Repeating unit of same monomer

Question 6

(e)

1. Linear polymers:

These polymers are formed of long straight chains. They can be depicted as:



For e.g., high density polythene (HDP), polyvinyl chloride, etc.

2. Branched chain polymers:

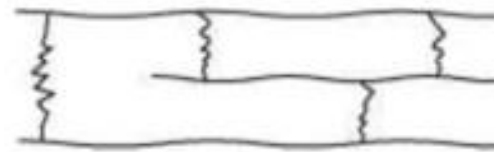
These polymers are basically linear chain polymers with some branches. These polymers are represented as:



For e.g., low density polythene (LDP), amylopectin, etc.

3. Cross-linked or Network polymers:

These polymers have many cross-linking bonds that give rise to a network-like structure. These polymers contain bi-functional and tri-functional monomers and strong covalent bonds between various linear polymer chains. Examples of such polymers include bakelite and melmac.



Question 6

(f)

2,4-DNP Test for aldehydes and ketones

Procedure: The 2,4-dinitrophenylhydrazine reagent (DNP reagent) and ethanol is added to the sample and shaken vigorously. If no precipitate forms immediately, allow the solution to stand for 15 minutes

Positive test: The formation of a yellowish precipitate shows the presence of aldehydes or ketones

OR

Tollen's test for aldehydes

Procedure: Freshly prepared Tollen's reagent is added to the sample (gentle heating can be employed if no reaction is immediately observed)

Positive test: Formation of silver mirror or a black precipitate is a positive test

Thank You